GATE 2027

Data Science & Artificial Intelligence (DA)

PREDICTED MOCK PAPER — 1

Real GATE level

Duration: 3 hours **Maximum marks: 100** **65 questions**

General instructions

1. The paper has two sections: **General Aptitude (GA)** — Q.1 to Q.10, 15 marks — and the **core Data Science & AI** section — Q.11 to Q.65, 85 marks.
2. Q.1–Q.5 and Q.11–Q.35 carry **one mark** each. Q.6–Q.10 and Q.36–Q.65 carry **two marks** each.
3. Each question is of one of three types, marked next to the question number:

Type	Description	Negative marking
MCQ	four choices, exactly one is correct	$-1/3$ (1-mark), $-2/3$ (2-mark)
MSQ	four choices, one or more are correct; no partial credit	none
NAT	a numerical value, to be entered directly	none

4. Statistical tables for the standard normal, t and χ^2 distributions, where needed, are the same as those supplied in the accompanying question bank.
5. The answer key appears at the end of this booklet. **Do not consult it until the three hours are over.**

How this paper was constructed. The subject-wise mark split, the MCQ/MSQ/NAT mix and the one-mark/two-mark balance all follow the average of the GATE DA 2024, 2025 and 2026 papers. Paper 1 mirrors the observed GATE DA difficulty curve; roughly 30% of the questions are direct, 50% are at true exam level and 20% are hard. See the accompanying *GATE DA 2027 Prediction & Pattern Report* for the full reasoning.

General Aptitude (Q.1–Q.10)

Q.1 – Q.5 carry one mark each.

Q.1 [MCQ • 1 Mark]

Select the word that is most nearly **opposite** in meaning to the word in capitals.

OPAQUE

(A) Dense (B) Obscure (C) Rigid (D) Transparent

Q.2 [MCQ • 1 Mark]

Select the most appropriate word to complete the sentence.

Despite the heavy rain, the match _____ as scheduled.

(A) preceded (B) proceeded (C) precedes (D) proceeding

Q.3 [NAT • 1 Mark]

The average of five numbers is 20. If the number 40 is removed from the set, the average of the remaining four numbers is _____.

Q.4 [MCQ • 1 Mark]

The angle between the hour hand and the minute hand of a correctly working clock at 3:30 is

(A) 75° (B) 80° (C) 90° (D) 60°

Q.5 [MCQ • 1 Mark]

Find the next term of the sequence 2, 6, 12, 20, 30, _____

(A) 44 (B) 40 (C) 36 (D) 42

Q.6 – Q.10 carry two marks each.

Q.6 [MCQ • 2 Marks]

The price of a commodity is first increased by 20% and then decreased by 20%. The net change in the price is

(A) a decrease of 2% (B) no change (C) a decrease of 4% (D) an increase of 4%

Q.7 [NAT • 2 Marks]

A alone can finish a piece of work in 12 days and B alone in 18 days. Working together, they finish it in _____ days (rounded off to one decimal place).

Q.8 [MSQ • 2 Marks]

In a class of 100 students, 60 play cricket, 45 play football and 20 play both. Which of the following statement(s) is/are correct?

(A) 40 students play only cricket

(B) 85 students play at least one of the two games

(C) 15 students play neither game

(D) 30 students play only football

Q.9 [MCQ • 2 Marks]

A train 200 m long crosses a stationary pole in 10 seconds. The speed of the train is

(A) 72 km/h (B) 20 km/h (C) 36 km/h (D) 120 km/h

Q.10 [MSQ • 2 Marks]

The daily sales (in units) of a shop from Monday to Friday are 20, 30, 25, 35, 40. Which of the following statement(s) is/are correct?

(A) The data set has a unique mode

(B) The median of the sales is 30

(C) The range of the sales is 20

(D) The mean of the sales is 30

Data Science & Artificial Intelligence (Q.11–Q.65)

Q.11 – Q.35 carry one mark each.

Q.11 [MCQ • 1 Mark]

If $X \sim \text{Poisson}(7)$, then $\text{Var}(X)$ equals

(A) 49 (B) $\sqrt{7}$ (C) 7 (D) 14

Q.12 [MCQ • 1 Mark]

The number of edges in a complete undirected graph on 8 vertices is

(A) 64 (B) 28 (C) 36 (D) 56

Q.13 [MSQ • 1 Mark]

Let F be the cumulative distribution function of a real-valued random variable. Which of the following statement(s) is/are **always** true?

(A) F is non-decreasing

(B) F is right-continuous

(C) F is continuous at every point of $\mathbb{R}$

(D) $\lim_{x \to -\infty} F(x) = 0$

Q.14 [MCQ • 1 Mark]

A superkey that contains no proper subset which is itself a superkey is called

(A) a candidate key (B) a foreign key (C) a partial key (D) a surrogate key

Q.15 [MCQ • 1 Mark]

If $X \sim \text{Binomial}\left(6, \frac{1}{3}\right)$, then $E[X]$ equals

(A) 2 (B) 6 (C) $\frac{4}{3}$ (D) $\frac{1}{3}$

Q.16 [MCQ • 1 Mark]

In a B^+ tree index, the actual data records (or pointers to them) are stored

(A) only at the leaf nodes (B) only at the root node

(C) at alternate levels of the tree (D) at every internal node

Q.17 [MCQ • 1 Mark]

Which of the following is an **unsupervised** learning algorithm?

(A) k-means clustering (B) Logistic regression (C) Random forest classifier (D) Support vector machine

Q.18 [MCQ • 1 Mark]

The worst-case time complexity of binary search on a sorted array of n elements is

(A) $\Theta(n \log n)$ (B) $\Theta(n)$ (C) $\Theta(1)$ (D) $\Theta(\log n)$

Q.19 [MCQ • 1 Mark]

Which statement about a real symmetric matrix A is correct?

(A) All its eigenvalues are positive (B) All its eigenvalues are real

(C) It is always invertible (D) Its eigenvalues always occur in complex conjugate pairs

Q.20 [MSQ • 1 Mark]

Which of the following sorting algorithms is/are **stable** in their standard implementation?

(A) Merge sort (B) Bubble sort (C) Heap sort (D) Selection sort

Q.21 [MCQ • 1 Mark]

A model that has high bias and low variance is best described as

(A) overfitting the training data (B) suffering from data leakage

(C) perfectly fitted (D) underfitting the training data

Q.22 [NAT • 1 Mark]

For the relation $R(A, B, C, D)$ with functional dependencies $A \to B$ and $B \to C$, the number of attributes in the closure A^+ is _____.

Q.23 [MCQ • 1 Mark]

$\frac{d}{dx}(x \ln x)$ equals

(A) $x \ln x + 1$ (B) $\ln x$ (C) $\frac{1}{x}$ (D) $\ln x + 1$

Q.24 [MCQ • 1 Mark]

The rank of the matrix $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ is

(A) 0 (B) 1 (C) 2 (D) 4

Q.25 [NAT • 1 Mark]

A binary classifier produces the confusion matrix $TP = 40$, $FP = 10$, $FN = 20$, $TN = 30$. Its precision is _____ (rounded off to two decimal places).

Q.26 [NAT • 1 Mark]

For two events, $P(A) = 0.6$, $P(B) = 0.5$ and $P(A \cup B) = 0.8$. Then $P(A \cap B) =$ _____ (rounded off to one decimal place).

Q.27 [NAT • 1 Mark]

Consider the Python code below.

```
a = [1, 2, 3]
b = a
b.append(4)
print(len(a))
```

The value printed is _____.

Q.28 [MSQ • 1 Mark]

Which of the following statement(s) about normal forms is/are correct?

(A) Every relation in BCNF is also in 3NF

(B) 3NF removes every kind of redundancy from a relation

(C) A relation in 3NF may violate 2NF

(D) Every relation in 3NF is also in 2NF

Q.29 [MCQ • 1 Mark]

A 3×3 matrix has $\det(A) = 12$ and two of its eigenvalues are 1 and 2. The third eigenvalue is

(A) 9 (B) 4 (C) 6 (D) 12

Q.30 [MCQ • 1 Mark]

For an exponential random variable with rate λ, the median equals

(A) $\frac{1}{\lambda}$ (B) $\frac{\ln 2}{\lambda}$ (C) $\frac{1}{\lambda^2}$ (D) $\frac{\lambda}{2}$

Q.31 [MSQ • 1 Mark]

Which of the following technique(s) can drive some model weights **exactly** to zero?

(A) Batch normalization
(B) L_2 (ridge) regularization
(C) L_1 (lasso) regularization
(D) Elastic-net regularization

Q.32 [NAT • 1 Mark]

A search tree has branching factor 3 and every goal lies exactly at depth 2 (the root is at depth 0). The number of nodes at depth 2 is _____.

Q.33 [NAT • 1 Mark]

The determinant of $\begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix}$ is _____.

Q.34 [MCQ • 1 Mark]

For a search tree in which every edge has the same positive cost, which strategy is guaranteed to return an optimal solution?

(A) Greedy best-first search (B) Depth-first search (C) Hill climbing (D) Breadth-first search

Q.35 [MCQ • 1 Mark]

The value of $\int_0^1 3x^2\,dx$ is

(A) 1 (B) $\frac{1}{3}$ (C) 3 (D) $\frac{3}{2}$

Q.36 – Q.65 carry two marks each.

Q.36 [MSQ • 2 Marks]

Which of the following statement(s) about k-fold cross-validation is/are correct?

(A) Increasing k generally reduces the bias of the performance estimate

(B) Every observation is used for validation exactly once

(C) Cross-validation removes the need for a separate test set

(D) Setting $k = n$ gives leave-one-out cross-validation

Q.37 [MSQ • 2 Marks]

Which of the following statement(s) about a foreign key is/are correct?

(A) A foreign key may reference the same relation in which it is defined

(B) A foreign key must be unique within the referencing relation

(C) A foreign key value may be NULL

(D) A foreign key must reference a candidate key

Q.38 [MCQ • 2 Marks]

A fully connected feed-forward network has 10 input units, one hidden layer of 20 units and 5 output units. Every layer except the input layer has a bias term. The total number of trainable parameters is

(A) 425 (B) 325 (C) 305 (D) 300

Q.39 [MSQ • 2 Marks]

Which of the following algorithms has/have $O(n \log n)$ **worst-case** running time?

(A) Quick sort with the first element as pivot (B) Insertion sort

(C) Heap sort (D) Merge sort

Q.40 [MSQ • 2 Marks]

For $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$, which of the following statement(s) is/are correct?

(A) The nullity of A is 1 (B) $\det(A) = 0$

(C) $Ax = 0$ has only the trivial solution (D) $\text{rank}(A) = 2$

Q.41 [MCQ • 2 Marks]

In the best case, alpha–beta pruning reduces the number of leaf nodes examined by minimax on a tree of branching factor b and depth d to

(A) $O\left(b^{d}\right)$ (B) $O\left(d^{b}\right)$ (C) $O(bd)$ (D) $O\left(b^{d/2}\right)$

Q.42 [NAT • 2 Marks]

PCA on a data set yields covariance eigenvalues 6, 3 and 1. The fraction of the total variance explained by the first two principal components is _____ (rounded off to two decimal places).

Q.43 [NAT • 2 Marks]

Consider the recursive function below.

```
def f(n):
if n <= 1:
return n
return f(n-1) + f(n-2)
```

The total number of invocations of `f` (including the initial call) when `f(5)` is evaluated is _____.

Q.44 [NAT • 2 Marks]

A system consists of three components that fail independently, each working with probability 0.9. The system functions if at least two of the three components work. The probability that the system functions is _____ (rounded off to three decimal places).

Q.45 [MSQ • 2 Marks]

Let A be a real symmetric $n \times n$ matrix. Which of the following statement(s) is/are correct?

(A) Eigenvectors corresponding to distinct eigenvalues are orthogonal

(B) A is orthogonally diagonalizable

(C) All eigenvalues of A are real

(D) A is always positive definite

Q.46 [NAT • 2 Marks]

Let $T(n) = 2T(n/2) + n$ with $T(1) = 1$, defined for n a power of 2. The value of $T(8)$ is _____.

Q.47 [MCQ • 2 Marks]

The keys 50, 30, 70, 20, 40, 60, 80 are inserted, in that order, into an initially empty binary search tree. The number of leaf nodes in the resulting tree is

(A) 4 (B) 2 (C) 5 (D) 3

Q.48 [NAT • 2 Marks]

A two-ply game tree has a MAX node at the root with three MIN children. The leaf utilities under the three MIN nodes are $(3, 5, 2)$, $(9, 1, 6)$ and $(4, 8, 7)$ respectively. The minimax value at the root is _____.

Q.49 [NAT • 2 Marks]

A table `Employee(eid, salary)` contains exactly five rows whose `salary` values are 10, 20, 20, 30, 40. The number of rows returned by

```
SELECT COUNT(DISTINCT salary)
FROM Employee
WHERE salary > 15;
```

is 1; the **value** returned by this query is _____.

Q.50 [MCQ • 2 Marks]

A hash table with 7 slots uses $h(k) = k \bmod 7$ with linear probing. The keys 10, 22, 31, 4, 15 are inserted in that order. The index at which key 4 is finally stored is

(A) 5 (B) 4 (C) 6 (D) 3

Q.51 [MCQ • 2 Marks]

A simple linear regression $y = \beta_0 + \beta_1 x$ is fitted by least squares to the points $(1, 2), (2, 3), (3, 5)$. The value of $\hat{\beta}_1$ is

(A) 0.75 (B) 1.0 (C) 1.5 (D) 2.0

Q.52 [MCQ • 2 Marks]

A file stores 10,000 fixed-length records of 100 bytes each. The block size is 1024 bytes and records

are *not* allowed to span blocks. The number of blocks required is

(A) 1000 (B) 977 (C) 10000 (D) 100

Q.53 [MCQ • 2 Marks]

In a production line 2% of the items are defective. An automated inspector flags a defective item with probability 0.95 and flags a non-defective item with probability 0.03. Given that an item has been flagged, the probability that it is actually defective is closest to

(A) 0.607 (B) 0.950 (C) 0.020 (D) 0.393

Q.54 [MSQ • 2 Marks]

Which of the following functions is/are convex on the whole of $\mathbb{R}$?

(A) $f(x) = x^3$ (B) $f(x) = |x|$ (C) $f(x) = x^2$ (D) $f(x) = e^x$

Q.55 [MCQ • 2 Marks]

Let $X \sim N(0, 1)$ and $Y = 2X + 3$. Then $P(Y > 5)$ is closest to

(A) 0.1587 (B) 0.8413 (C) 0.0228 (D) 0.3085

Q.56 [NAT • 2 Marks]

Gradient descent is applied to $f(x) = (x - 3)^2$ with learning rate 0.25, starting from $x_0 = 0$. The value of x_1 after one update is _____ (rounded off to two decimal places).

Q.57 [NAT • 2 Marks]

A random sample of size 9 from a normal population gives $\bar{x} = 15$ and $s = 3$. The margin of error of the 95% confidence interval for μ is _____ (rounded off to three decimal places). *(Use $t_{0.025,\,8} = 2.306$.)*

Q.58 [MCQ • 2 Marks]

A weighted undirected graph has vertices $\{A, B, C, D, E\}$ and edges with weights $AB = 4$, $AC = 2$, $BC = 1$, $BD = 5$, $CD = 8$, $CE = 10$, $DE = 2$. The length of the shortest path from A to E is

(A) 12 (B) 11 (C) 10 (D) 9

Q.59 [MSQ • 2 Marks]

Two random variables satisfy $\mathrm{Var}(X) = 4$, $\mathrm{Var}(Y) = 1$ and $\mathrm{Cov}(X, Y) = 1$. Which of the following statement(s) is/are correct?

(A) The correlation between X and Y is 0.5 (B) $\mathrm{Var}(X - Y) = 3$

(C) $\mathrm{Var}(X + Y) = 7$ (D) X and Y are independent

Q.60 [NAT • 2 Marks]

For the confusion matrix $TP = 40$, $FP = 10$, $FN = 20$, $TN = 30$, the F_1 score is _____

(rounded off to three decimal places).

Q.61 [MCQ • 2 Marks]

The relation $R(A, B, C, D)$ has functional dependencies $A \to B$, $B \to C$ and $C \to D$. The highest normal form satisfied by R is

(A) BCNF (B) 2NF (C) 1NF only (D) 3NF

Q.62 [MCQ • 2 Marks]

The largest eigenvalue of $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ is

(A) 4 (B) 10 (C) 7 (D) 5

Q.63 [NAT • 2 Marks]

A data set has covariance matrix $\begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}$. The variance captured by the first principal component is _____.

Q.64 [MCQ • 2 Marks]

The function $f(x) = x^3 - 3x$ attains a local minimum whose value is

(A) 2 (B) -1 (C) -2 (D) 0

Q.65 [MSQ • 2 Marks]

For the 8-puzzle, which of the following heuristic function(s) is/are **admissible**?

(A) The constant function $h(n) = 0$

(B) Sum of Manhattan distances of the tiles from their goal positions

(C) Twice the sum of the Manhattan distances

(D) Number of misplaced tiles

Answer Key

Answers only — no worked solutions. For NAT questions any value within the usual GATE tolerance of the figure shown is accepted. The **Subject** column tells you which topic to revise when you get one wrong. A key with several letters (e.g. **ABD**) is an MSQ.

Q	Ans	Subject	Q	Ans	Subject	Q	Ans	Subject
1	D	GA	**2**	B	GA	**3**	15	GA
4	A	GA	**5**	D	GA	**6**	C	GA
7	7.2	GA	**8**	ABC	GA	**9**	A	GA
10	BCD	GA	**11**	C	PS	**12**	B	PDSA
13	ABD	PS	**14**	A	DB	**15**	A	PS
16	A	DB	**17**	A	ML	**18**	D	PDSA
19	B	LA	**20**	AB	PDSA	**21**	D	ML
22	3	DB	**23**	D	CO	**24**	B	LA
25	0.80	ML	**26**	0.3	PS	**27**	4	PDSA
28	AD	DB	**29**	C	LA	**30**	B	PS
31	CD	ML	**32**	9	AI	**33**	8	LA
34	D	AI	**35**	A	CO	**36**	ABD	ML
37	ACD	DB	**38**	B	ML	**39**	CD	PDSA
40	ABD	LA	**41**	D	AI	**42**	0.90	ML
43	15	PDSA	**44**	0.972	PS	**45**	ABC	LA
46	32	PDSA	**47**	A	PDSA	**48**	4	AI
49	3	DB	**50**	A	PDSA	**51**	C	ML
52	A	DB	**53**	D	PS	**54**	BCD	CO
55	A	PS	**56**	1.50	CO	**57**	2.306	PS
58	C	PDSA	**59**	ABC	PS	**60**	0.727	ML
61	B	DB	**62**	D	LA	**63**	6	LA
64	C	CO	**65**	ABD	AI			

Score sheet

Fill this in after checking the key. Apply GATE negative marking: $-1/3$ for a wrong 1-mark MCQ and $-2/3$ for a wrong 2-mark MCQ; MSQ and NAT carry no penalty.

Subject	Questions	Marks available	Marks scored	Time taken
General Aptitude	10	15		
Probability & Statistics	10	15		
Linear Algebra	8	12		
Calculus & Optimization	5	8		
Programming, Data Structures & Algorithms	10	16		
Database Management & Warehousing	8	12		
Machine Learning	9	14		
Artificial Intelligence	5	8		
Total	**65**	**100**		